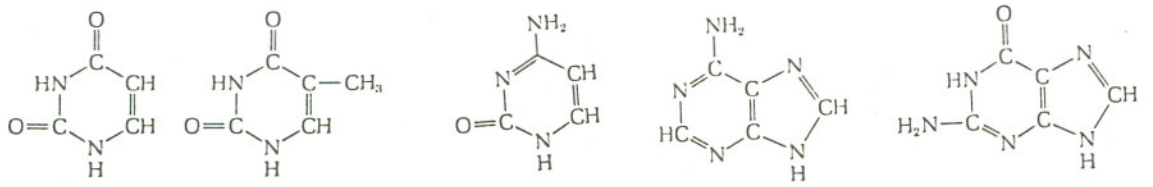
BIOCHEMISTRY CUMULATIVE EXAM

November 16, 1991

Macromolecular Structure

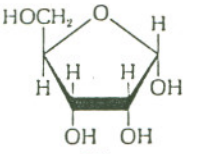
1) Depicted below are the five nucleosides commonly found in DNA and RNA. Give the correct name for each. (5 points)



1 2 3 4 5

2) Draw the two Watson-Crick base pairs found in DNA. Be sure to show the hydrogen bonds. (8 points)

3) Shown below is ribose, the sugar found in RNA. Indicate which two hydroxyl groups form the phosphodiester bonds. Indicate which hydroxyl is missing in deoxyribose. (6 points)



α-D-Ribose

4) There are two common types of DNA double helix known that correspond more or less to the original model proposed by Watson and Crick. These are usually called A and B DNA and differ in helical pitch and other (relatively) minor details. Another, radically different, form of double helix is known to be possible *in vitro*. It may be important *in vivo* in gene regulation and other such processes although definitive proof does not exist. What is this other form of DNA double helix called (2 points) and how does it principally differ from A and B DNA (4 points)?

5) In reading or speaking of protein structure one often comes across the terms primary, secondary, tertiary and quaternary structure. Describe briefly what is meant by each of these terms. (8 points)

6) Depicted on the next page are the 20 amino acids ordinarily found in proteins. Give the name of each. (20 points)

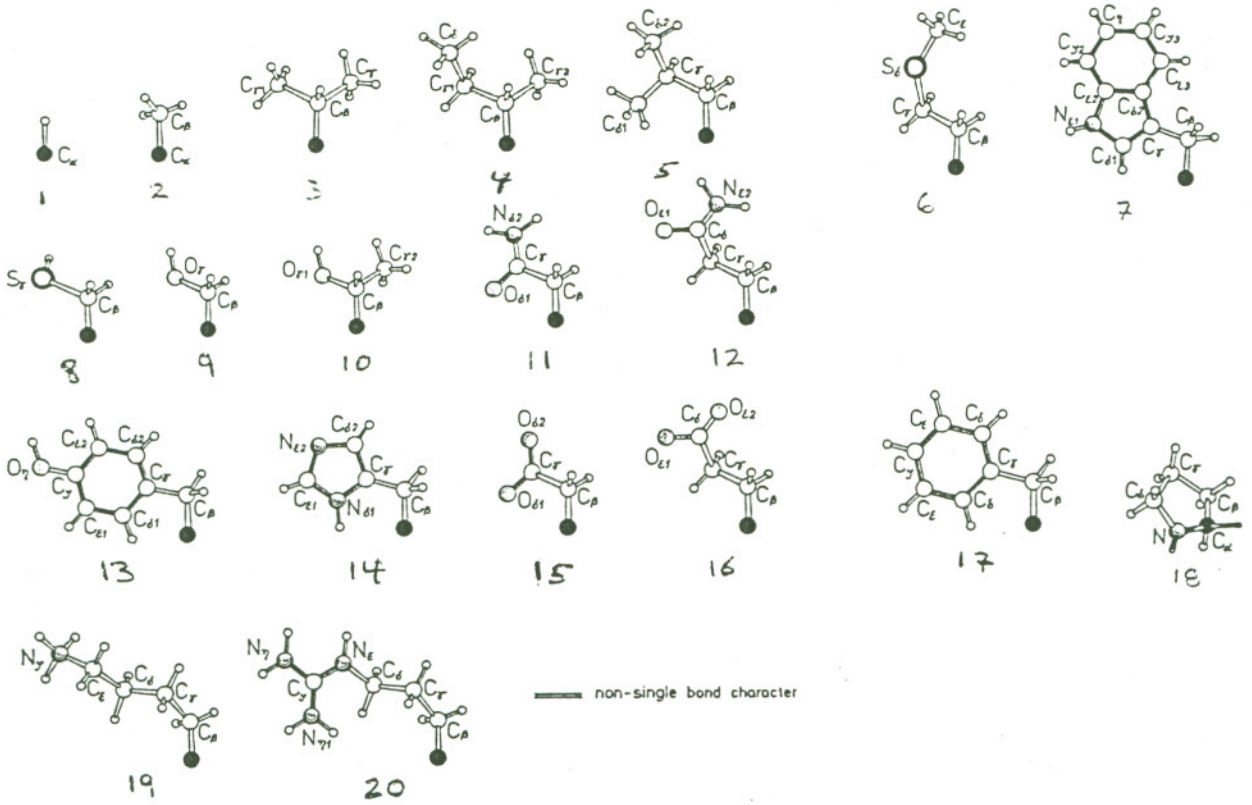


Figure 1-1. The 20 standard amino acid side chains. For part of the main chain is inserted. All other side chains are shown as they emerge from the C_α-atom in the main chain. All C_α-atoms are black.

7) One of the amino acids shown above has the ability to form disulfide bonds. The formation of a disulfide bond in model systems is worth only a few kcals/mol. However, the formation of disulfide bond in a protein may stabilize its structure by tens of kcals/mol. Explain this apparent contradiction. Specifically, why does a disulfide often do so much to stabilize a protein structure? (20 points)

8) Despite the fact that disulfide bonds are stabilizing to protein structures many proteins do not have them. Indeed, disulfides seem limited to proteins that are secreted from the cell. Give a rationale for this. (5 points)

9) Beta sheet is commonly found structural motif in proteins. Two kinds of beta sheet are known. Draw an example of each kind of beta sheet illustrating in particular how the peptide backbone is arranged. Because of the focus on the arrangement of the backbone use polyglycine in your drawing. Show three strands in each sheet, each strand with three amino acids. Do not show hydrogens except for those participating in hydrogen bonds. Be sure to show hydrogen bonds. (20 points) Below each illustration write the name of that particular sort of beta sheet. Give the name even if you cannot draw the structure. (2 points)